Zenith: Learning to Reason About Financial Change

Method and development evidence

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Abstract

Financial language can describe substantial growth while revealing a weaker outlook than an earlier disclosure. A useful reasoner must recover that comparison, identify its economic channel and distinguish it from a return still available after entry. Zenith develops this capability through dated evidence, an inspectable change representation and feedback on subsequent outcomes. Its implemented development route compares direct reasoning with an interpretation–forecast graph, and separates reasoning from numerical calibration. On 33 previously examined A-share development dates, two fixed statistical controls had positive outcome alignment but excessive forecast magnitudes. Scaling fitted only on earlier matured predictions reduced their mean squared errors below a zero-relative-return reference by approximately 0.7–0.8%. This modest, design-exposed result does not establish an LLM advantage. Four newly compiled guidance-update cases, drawn from six original issuer releases, supply richer comparisons, including mixed revisions and unchanged revenue guidance alongside higher margins. The model comparison and learned reasoning experiment remain unexecuted. The research hypothesis is that feedback improves the recognition and use of financial changes enough to improve later decisions beyond fixed reasoning, retrieval and calibration.

1 The financial reasoning problem

Salesforce's May 2023 release left fiscal-2024 revenue guidance at $34.5–34.7 billion while raising non-GAAP operating-margin guidance from 27% to 28% [16, 17]. A representation that reports only positive earnings language misses the distinction between a revenue revision and a profitability revision. Adobe's June 2023 release requires another distinction: some annual targets rose while Digital Experience revenue targets fell relative to their last published ranges [1–3]. These are observable changes in issuer guidance. Neither example supplies analyst consensus or establishes an unexpected improvement from the market's perspective.

Even a correct comparison leaves a second question: what return remains available at the intended entry? A release after the close may affect the next opening price. A forecast for buying at that open must concern subsequent continuation, reversal or delayed

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transmission. Work on headline-based return prediction distinguishes price response from later predictability [13]; Zenith treats the remaining effect as something to measure. Economic links can motivate indirect transmission [8], but a shared name, news tag or semantic similarity does not establish such a link.

The proposed learning contribution is specific: experience should improve the system's representation and use of financially meaningful changes, with later decision gains beyond a strong fixed reasoner and numerical adaptation. This claim is untested. The work reported here establishes an executable development route, a measured calibration diagnosis and a concrete source-rich replacement task.

2 Separating interpretation from payoff

Let E_t contain evidence available by the decision time and X_t contain admissible numerical state. Zenith's graph produces an interpretation $G_t = R_\theta(E_t, X_t)$. Its current contract asks for supported facts, change relative to earlier evidence, an economic channel, what may already be priced, and unresolved questions. References must identify supplied evidence. The forecast node receives both G_t and the original evidence, so a compressed interpretation cannot silently become its only source.

The intended numerical interface is

$$\hat{y}_t = b_\phi(X_t) + h_\psi(G_t, X_t), \tag{1}$$

with the baseline and incremental map fitted from earlier matured examples. This makes two failures distinguishable: a poor representation and an excessive payoff assigned to a useful representation. The present adapter implements a small special case: an economic-change signal in $[-1, 1]$, one chronologically fitted nonnegative coefficient capped at 0.05, and a zero baseline. An unknown signal remains null in the recorded output; the numerical system applies a declared zero correction. A scalar cannot preserve every dimension of a mixed event. The richer textual representation remains available to the forecast node and for inspection.

A strong direct comparator receives the same evidence, numerical state, decision clock and target, with instructions to perform the relevant comparisons itself. The graph adds an intermediate representation and a second call. Graph composition is established prior art [6]; its additional cost must earn a measured benefit. This initial comparison is matched on inputs and requested model/provider, not on total tokens or calls.

Temporal price–text conditioning and financial instruction tuning appear in Yu et al. [21]. Zenith adopts the need for dated mixed evidence; it does not reproduce their return bins, fine-tuning or reported results. Huang et al. [11] distinguish retrieving candidate cross-stock relations from judging them against disclosure evidence. This motivates separating issuer relevance from a hypothesized indirect channel. Their graph construction and portfolio rule are not imported. These mechanisms require source-grounded review: valid JSON with valid reference identifiers does not prove that its claims follow from the text.

3 What the existing forecasts reveal

The A-share development panel uses ten issuers and 41 admitted issue dates in 2023 from the pinned TraderHarness archive [5, 10]. After eight initial fit dates, expanding chronological controls produce 330 forecasts across 33 dates. The numerical control uses trailing market state, evidence metadata and issuer identity. A matched text control adds character 2–4-gram TF–IDF with at most 8,192 features. Transformations and ridge coefficients use earlier matured training outcomes; the ridge penalty is one. This modest target-trained text comparison is motivated by Ke et al. [12], without reproducing their estimator. It uses scikit-learn [14].

The target is gross stock wealth return from the next open through the fifth session's close, minus the matching CSI 300 price-index return. It is a forecast target, not the return of an executable hedged portfolio. Each date has equal weight; within a date, each admitted issuer has equal weight. For forecast q and outcome y, the square-loss identity

$$\mathrm{MSE}(q) - \mathrm{MSE}(0) = \mathbb{E}[q^2] - 2\mathbb{E}[yq] \tag{2}$$

separates forecast energy from outcome alignment. The numerical control has energy 0.00048579 and alignment 0.00008124; the text control has energy 0.00056507 and alignment 0.00008840. Both align positively with outcomes, but their magnitudes more than exhaust that alignment. The best unconstrained scalar fitted retrospectively on all 33 dates is 0.1672 for the numerical control and 0.1564 for text. These all-date fits diagnose overscaling; they are not later-example predictions.

We therefore measured a chronological scaling rule. At each query date, it fits one coefficient from complete earlier date groups whose outcomes have already matured. With $n \geq 4$ such dates,

$$c_t = \mathrm{clip}_{[0,1]} \frac{\mathbb{E}_{<t}[yq]}{\mathbb{E}_{<t}[q^2](1 + 4/n)}. \tag{3}$$

Otherwise, or when the denominator is zero, $c_t = 0$. The factor $4/n$ supplies a zero-centred penalty equivalent to four dates of observed signal energy. Warm-up forecasts remain in the score as zero corrections. This calibration design followed inspection of the original aggregate errors. It was not retuned after observing the new results, but the whole comparison is exposed development evidence, not an untouched assessment. Selection on evaluation results is itself a source of optimism [7].

Control	Original MSE	Chronologically scaled MSE
Zero relative return	0.00109518	0.00109518
Numerical, metadata, issuer	0.00141850	0.00108729
Same inputs plus text	0.00148344	0.00108654

Table 1: Actual development results on identical 33-date support. No LLM, learned-reasoning or final-assessment result is represented here.

Relative to the numerical control, the raw text increment has energy 0.00008824 and residual alignment 0.00001165; it also overshoots. Applying the same chronological rule

to that increment on top of the scaled numerical baseline gives MSE 0.00108653. Its near equality with the scaled text control supplies no compelling evidence of a distinct text advantage. The approximately 0.7–0.8% reduction below zero is small, dependent on the development design and not accompanied by a significance claim. It changes the diagnosis from no usable alignment to positive sample alignment overwhelmed by magnitude. It neither tests nor rejects the unexecuted LLM.

4 Giving the reasoner the deciding evidence

The A-share archive contains short headlines and announcement titles, often without the earlier disclosure needed for a comparison. A bounded retrieval patch added only five earlier comparison items to training inputs. Some issuer tags refer to related listed companies rather than the forecast issuer. These limits motivate a fuller-source recurring event task before optimization.

The replacement development set contains four guidance updates from Adobe and Salesforce, supported by six original issuer releases. They were chosen purposively for accessible current/prior disclosures before scoring their returns. Table 2 records source comparisons, not model answers. The compiler checks 13 metric comparisons across these cases and keeps their comparison oracle separate from the model request. It preserves range endpoints and widths: an unchanged midpoint need not imply unchanged guidance.

Update	Supported change in annual guidance
Adobe, 15 March 2023	Net new Digital Media ARR rises from \$1.65bn to \$1.70bn; non-GAAP EPS rises from \$15.15–15.45 to \$15.30–15.60. Annual revenue guidance is omitted from this update [1, 3].
Adobe, 15 June 2023	Revenue midpoint rises from \$19.20bn to \$19.30bn, but Digital Experience revenue midpoint falls from \$4.975bn to \$4.875bn. Those metrics compare with December; EPS and ARR compare with March [1–3].
Salesforce, 31 May 2023	Revenue stays at \$34.5–34.7bn; non-GAAP operating margin rises from 27% to 28%, and EPS from \$7.12–7.14 to \$7.41–7.43 [16, 17].
Salesforce, 30 August 2023	Revenue rises to \$34.7–34.8bn; non-GAAP operating margin to 30%, and EPS to \$8.04–8.06 [15, 17].

Table 2: A source-feasibility cohort, not a representative opportunity stream. Fiscal period and accounting basis are matched. An omitted metric is not automatically reaffirmed or withdrawn. ARR and operating-margin point targets are approximate, as stated by the issuers.

Adobe target sections are extracted from retained PDFs. Salesforce inputs contain selected annual-guidance values transcribed from visually reviewed tables; other measures and reconciliation tables are not supplied to the model. Current-site navigation and later editorial additions are excluded. This tests reasoning over reviewed extracts, not autonomous full-report extraction. SEC acceptance records cross-check the release clocks. The reconstructed decision time is acceptance plus five minutes, with entry at

the next regular open. Retrieved issuer mirrors are not authenticated historical snapshots, and exact identity with historical SEC exhibit bytes has not been proved. These distinctions remain in the source manifest.

Price support comes from the pinned FinMultiTime archive [20]. Twenty completed sessions and five future sessions exist for each issuer and SPY, with exchange holidays respected. The historical price adjustment basis is unresolved, including distribution treatment. Consequently the current task scores no returns. It asks whether the reasoner recovers financial changes while retaining unknown expectations and mixed directions. Four requests are compiled and held; none has a model answer. The price gap does not prevent this representation test, but prevents a credible trading result from these cases.

5 Developing a base, then testing learning

The next A-share batch contains an excluded compatibility date and six development dates selected from their inputs. Each date requires one direct call, one change-interpretation call and one graph-forecast call: at most 21 calls in total. The pilot verifies the actual contracts before comparison. Rejected responses and provider failures remain recorded. Inspection will ask whether outputs identify the right issuer and period, preserve revisions and uncertainty, and reason about the effect remaining after entry. Direct and graph forecasts receive the same chronological calibration rule; post-warm-up comparisons also report the reference methods on exactly the same dates. Six dates can diagnose behavior, not establish superiority.

Development may alter retrieval, prompts, tools, topology, model, event family or horizon. Each materially different version retains its inputs, attempts and design exposure. A frozen comparison fixes choices for that run; it does not permanently restrict the product to two nodes or a single mutable field. Future assessment must report its role in development honestly. Modern models may remember historical events, so chronological input filtering alone does not establish freedom from training contamination [9].

Once the base demonstrates useful reasoning, feedback can propose changes to recurring procedures: retrieve each metric's last publication, verify the accounting period, inspect a countervailing segment, or decline an unsupported surprise claim. Textual optimization and reflective search supply relevant mechanisms [4, 24]. Zenith already has native proposal paths; GEPA's adapter interface is a candidate for connecting reflective search to them. This hypothesis does not require a new general optimizer.

The decisive comparison holds numerical adaptation common between a fixed reasoner and one whose procedures learn. Retrieved experiences form another control, related to FinMem [22]; persistent investment guidance is related to FinCon [23]. Zenith's implemented memory retrieval combines forward-text similarity with numerical-state proximity. Wang and Yeung [19] also motivate numerical and narrative analogs, but their financial-network indicators are different targets. Their reported performance is not evidence for stock-return prediction here.

An outcome-only teacher and a teacher that additionally sees later reporting can test privileged feedback [18]. Both must produce procedures usable from later forward evidence. A compelling explanation of a past miss is not by itself a transferable lesson. Neither future-news text similarity nor output validation establishes explanation

fidelity. Learning is useful only if later behavior and decisions improve beyond the appropriate controls.

6 From forecasts to decisions

A trading claim needs a position-sizing rule, an executable reference asset, cash and overlapping positions, entry and exit rules, distribution entitlements, transaction costs and relevant market constraints. If positions are decided before the opening price is observable, the rule cannot use that price while assuming execution at the same price. Remaining returns must be measured after the declared information and execution delay.

The existing A-share label mixes stock wealth and index price returns and omits costs, taxes, lots and execution restrictions. The guidance cohort has an unresolved price basis. Neither supports a profitability claim. The current evidence supports a narrower conclusion: baseline forecasts can contain useful alignment obscured by scaling, and recurring disclosures supply testable financial distinctions absent from short titles. Whether LLM reasoning extracts those distinctions reliably, whether feedback improves their later use, and whether either changes feasible decisions remain empirical questions.

Reproduction and supporting evidence. The repository retains source pins, per-date measurements, compiled comparisons and exposure notes. The reproduction map in `paper/README.md` links these artifacts, the active commands, the spending boundary and implementation owners. An earlier 2015 manual-rule pilot established pipeline feasibility only. Its immutable results, provider accounting, conditional formal model and routing tests remain supporting artifacts, not evidence of learned financial reasoning.

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